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Diploma in Information Technology (DIT 36)

Seminar on Issues and Trends in Information Technology (ITEC304)

INSTRUCTIONS:

1. Read the assigned article: "Top 5 Uses of Autonomous Security Patrol Robots in 2025" by Help Force News (December 22, 2025).
2. Write a well-organized reaction paper addressing all questions below. Use complete sentences, provide examples or evidence from the article, and include your personal reflections or analysis.
3. Answer each question in the space provided.
4. Your responses should be original, properly formatted, and free of plagiarism. Be thoughtful and concise in your responses.

ASSIGNMENT 4 : Top 5 Uses of Autonomous Security Patrol Robots in 2025

1. Impact on Security Operations

Based on the article, how have autonomous security patrol robots transformed security operations in different industries? Discuss both their advantages (such as continuous monitoring, improved threat detection, reduced manpower costs, and full-area coverage) and challenges (such as high implementation cost, maintenance needs, and dependence on technology).

Various industries implementing autonomous security patrol robots have significantly enhanced security operations by offering continuous 24/7 monitoring and advanced threat detection capabilities in various settings such as malls, warehouses, and hospitals. These robots help lower manpower costs since a single unit can patrol extensive areas without resting. They also increase coverage by using cameras and AI to simultaneously monitor multiple locations at the same time. However, using such advanced technology requires maintenance so maintaining few units can be costly and complicated.

2. Public Safety and Experience

In what ways do autonomous security robots improve public safety and security awareness in places like malls, hospitals, and corporate campuses? Explain how features such as real-time surveillance, AI detection systems, and visible patrol presence influence how people feel about safety and security.

Autonomous security robots improve public safety by supplementing surveillance and allowing for quick identification of suspicious activity. Cameras and AI can help them monitor high traffic areas and raise alarms in emergencies. The fact that people can see them helps people feel safer knowing that the area is always under surveillance. They also help visitors in malls and hospitals by guiding them and watching entrances. These advantages make people feel more secure in settings where robotic security systems are deployed.

3. Technology, Accuracy, and Efficiency

Evaluate the effectiveness of autonomous security robots compared to human security personnel.

How do robots improve patrol efficiency, accuracy in detecting incidents, and consistency of security coverage? Also, identify possible limitations when robots are used in unpredictable or emergency situations.

Autonomous security robots enhance patrol efficiency by continuously monitoring large areas without fatigue, allowing for consistent security coverage that human personnel might struggle to maintain over long shifts. Their AI-powered systems improve accuracy in detecting incidents by quickly analyzing data from cameras and sensors to identify potential threats in real time. This reduces the chances of human error or oversight. However, robots can face limitations in

unpredictable or emergency situations where human intuition, judgment, and adaptability are crucial. Technical malfunctions or the inability to respond flexibly to complex scenarios may reduce their effectiveness in such cases.

4. Ethical and Workforce Considerations

According to the article, a robot can replace several human guard positions in certain settings. Discuss how this affects human security workers. Do robots fully replace human guards or act as support tools? Explain possible ethical concerns such as job displacement, reduced human employment, and changes in job roles.

Autonomous security robots have the potential to replace several human security positions in certain environments, which can significantly impact human security workers. Rather than fully replacing human guards, these robots often serve as support tools that enhance overall security efforts. However, there are ethical concerns to consider, such as job displacement and the reduction of human employment opportunities in the security industry. This shift may also change the nature of security jobs, requiring workers to adapt to new roles that focus more on managing and maintaining robotic systems rather than traditional patrol duties. Balancing the benefits of technology with the need to protect workers' livelihoods remains a key challenge.

5. Future of Coffee Shop Industries

Based on the article, how do you envision the future of security systems with the continued development of autonomous security robots? Do you think human security personnel will still be essential in the future? Justify your answer using ideas from the article.

With the ongoing development of autonomous security robots, security systems in coffee shops and similar industries are likely to become more advanced and efficient, offering continuous monitoring and quick threat detection. However, human security personnel will remain essential in the future. While robots can handle routine surveillance and data collection, humans provide critical judgment, empathy, and flexibility in handling complex or unexpected situations. The article suggests that robots complement human workers rather than replace them entirely, creating a partnership where both contribute to safer environments. This blend of technology and human oversight will likely define the future of security in coffee shops and beyond.